Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI technique that combines information retrieval with large language models (LLMs) to generate accurate, context-aware responses. Instead of relying only on the model's training data, RAG retrieves relevant information from external sources such as documents, databases, or knowledge bases before generating an answer.

How RAG Works

1. User Query: A user submits a question.
2. Retrieval: The query is converted into embeddings and searched against a vector database (e.g., FAISS, Pinecone, ChromaDB) to find the most relevant documents.
3. Context Augmentation: The retrieved content is added to the original query.
4. Generation: The LLM uses both the query and retrieved context to produce a more accurate and grounded response.

Key Components

- Knowledge Source: PDFs, websites, databases, manuals, or enterprise documents.
- Embedding Model: Converts text into vector representations.
- Vector Database: Stores embeddings and performs similarity search.
- Retriever: Identifies the most relevant documents.
- Large Language Model (LLM): Generates the final response using the retrieved context.